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Design of a Memory System for Sparse Data Processing

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THESIS

DESIGN OF A MEMORY SUB-SYSTEM FOR SPARSE DATA PROCESSING

August 6, 2025

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Résumé

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Abstract

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List of Publications

This thesis is based on the following publications:

- I Valentin Isaac--Chassande, Adrian Evans, Yves Durand, and Frédéric Rousseau. 2024. Dedicated Hardware Accelerators for Processing of Sparse Matrices and Vectors: A Survey. *ACM Trans. Archit. Code Optim.* 21, 2, Article 27 (Feb. 2024), 26 pages. Reference [42]
- II Valentin Isaac--Chassande, Adrian Evans, Yves Durand, and Frédéric Rousseau. 2024. SpDCache: Region-Based Reduction Cache for Outer-Product Sparse Matrix Kernels. In *2024 IEEE 35th International Conference on Application-specific Systems, Architectures and Processors (ASAP)*. IEEE Computer Society, Los Alamitos, CA, USA, 3–7. Reference [43]
- III **TODO**

The papers will be referred to in the thesis using their Roman numerals.

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Notations

$$\forall (a, b) \in \mathbb{Z}^2, \llbracket a, b \rrbracket = [a, b] \cap \mathbb{Z}$$

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Introduction

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From SoA paper

The execution time of modern scientific and engineering applications is dominated by the manipulation of large sparse matrices, *i.e.* matrices whose number of non-zero entries NNZ is much smaller than their dimensions $M \times N$. Matrices derived from computational physics problems are often sparse and regular because they derive from a relatively small number of differential terms. Algebraic graph problems produce matrices which are still sparse but usually less regular [16]. Similarly, the weight and activation matrices used by Convolutional Neural Networks (CNN) may hold millions of terms, but most of them are explicitly set to zero (*e.g. rectified*) without loss of information [28]. It was recognized early in the 1960s [71] that working on these sparse data sets using dense array representations was inefficient. The seminal work of OSKI [85] in 2003 formalized the sparse representation alternatives and identified many optimizations. This opened the path for specialized software libraries for sparse algebra [86]. However, the single-core performance of software implementations can be deceiving, often only reaching 10% of peak performance [28].

The main matrix multiplication kernels that are of interest are matrix-vector (MV) and matrix-matrix (MM), which include SpMV and SpMSpV ($A \times b = c$), SpGEMV ($A \times b + d = c$), SpMM and SpMSpM ($A \times B = C$), and SpGEMM ($A \times B + D = C$), with *Sp* for *sparse* and *GE* for *general*. SpMV, as well as its variant SpGEMV, are by far the dominant operations of modern algebraic kernels (linear solvers, eigensolvers) which have been explicitly based on them [25, 32, 37, 44, 49, 67, 89]. Since these kernels are ubiquitous, their efficient implementation is primordial. In contrast, the SpMSpM multiplication is less frequent in standard algebraic kernels, but it is extensively used in algebraic graph manipulation [4, 14, 17, 23, 31, 46, 68], as well as in multigrid solvers [3, 13, 53, 70]. The growing importance of large graphs has motivated new research on optimizing SpMSpM applications.

The two matrix operations, SpMV and SpMSpM, pose orthogonal computational challenges: (1) access to sparse input data, (2) *reduction* operations on the intermediate result and (3) formatting and writing back the result to memory. The different algorithms, which

are detailed in Section 2.1, are built on different articulations of these three aspects. The sparse data storage scheme used in memory dictates the first aspect and is briefly explored in Section 2.1.1. The third aspect, *i.e.* result storage, is sometimes overlooked when the output is dense and not too large. For example, when running on a conventional CPU which is memory-bound, it becomes crucial to streamline the write accesses in order to optimize cache usage when writing a large dense vector output. The second aspect, *i.e.* the *reduction* operations, deserves different solutions according to the execution platform. On a conventional CPU, the bottleneck is memory throughput. Thus, most software implementations of SpMV deal with *a priori* reorganization of the input matrices, by row ordering, blocking and loop optimization, for improving the data locality of operations. The perspective for SpMV is different when dealing with GPU platforms. The challenge shifts to finding a balanced assignment of the numerous threads and warps, and to the use of local memory resources [57]. SpMSPM is far more complex: the intermediate *reductions* introduce new *NNZ* values at random locations, which trigger costly memory allocations and list management operations [29]. As a consequence, this phase dominates the execution time and becomes the main memory bottleneck on a regular CPU. The issue is similar on GPUs because of the limited size of the scratchpad memory for processing large matrices, which requires complex memory management and load balancing [22, 63, 90].

The main motivation to use custom hardware accelerators is to make better use of the intermediate memory hierarchy, ensuring the input data is available near the cores and also off-loading the accumulation and *reduction* operations from the processors. The focus of this paper is to review these architectures, to compare them and to analyze their performance. Our main focus is on accelerators that eventually target custom or ASIC implementations, although we do touch on the contributions from accelerators that were prototyped on FPGAs. The reader is referred to [59] for a survey of acceleration techniques for CPUs, GPUs and FPGAs. Section 2.1 reviews the storage schemes and the three basic algorithms with a focus on the data access patterns and their impact on the memory hierarchy. In Section 2.2 we study how the different algorithms stress the memory systems and propose an analytical model based on these insights. In Section 3.2 we present the operation of several state-of-the-art hardware accelerators, in the light of these hardware challenges. In Section 3.3, we systematically compare these accelerators. Throughout this study, we try to be consistent, and we hope that our nomenclature will be useful for architects and designers who intend to implement dedicated hardware for problems with sparse datasets.

Chapter 1

Plan (Brouillon)

- Introduction (~ 3 p.)
 - "light" > give the entry point on sparse matrices and irregular accesses (1-2 p.)
 - Mains contributions (1 p.)
 - introduce the structure of the manuscript
- Background (~ 5 p.)
 - irregular accesses -> def, issues (in memory but not only?), domains/applications (including sparse matrix processing) (1 p.)
 - Focus on sparse matrices, context: it is widely used, in many domains, sparse, large, structures
 - Sparse matrices in a kernel: show where the accesses are regular vs irregular (infos on basic sparse formats (other sparse formats in appendix x), tiling, basic kernels, basic algos...) (2-3 p.)
 - Hardware challenges clearly defined (gather, scatter...) (1 p.)
- SoA (~ 11 p.)
 - Present solutions in SW for CPU/GPU, such as special sparse formats, possibility to play on algo (gust analysis), tiling methods adapted for sparse data (specific sparse format + specific algos) (expliquer clairement que ces solutions sont intéressantes et pourquoi on se concentre sur le HW) (1-2 p.)
 - Present solutions in HW (dedicated accelerators) -> comparison table along common characteristics (1-2 p.)
 - Presentation of the major accelerators (selection of 3 important: processor assist PHI, standalone GAMMA, multi algo SPADA, ACES) (others in appendix xi) (3-4 p.)
 - Problem of comparison / comparison of standalone / other analysis (issue of matrix structure) -> Objectives: convince reader that SoA was done in depth, and acknowledge the matrice structure issue (1 p.)
 - Takeaways -> paths for designing an accelerator (end of SoA paper) (1-2 p.)
 - We choose a path -> show which one (coarse grain, at the level of the SoA, comparable to SortCache, PHI...)
- High level architectural presentation of SpDCache (~ 10 p.)
 - Reminder of the path we choose with more details: cache / reductions / OP SpMV / ... (1 p.)
 - Emphasis the comparison between concerned SoA accelerators -> show what is already done and useful, show what is missing

- Present the case of banded matrices as an example of structure to leverage **(1-2 p.)**
- SpDCache, architecture presentation with two regions, details on REDs transactions, IBRC vs OBRT **(7 p.)**
- Initial results from Python model (ASAP) and note that a probabilistic model and RTL model will follow
- Mathematical analysis of the cache memory accesses for processing sparse matrices **(~23 p.)**
 - Explain the objectives of this chapter (~formal justification of the concept + cache size/region dimensioning) **(1 p.)**
 - Describe the design space of the path we choose (with reminders if necessary) -> introduce base parameters and/or variables, give the exact algorithm... **(1 p.)**
 - Show the interest of separate cache regions for A/b and c (formal) -> show that a GC is obviously not efficient when random accesses / confirm why SortCache, PHI, ... exist **(2 p.)**
 - Show that A and b are not a limiting factor with a formal reasoning -> give the needed space for GRC when prefetcher (clear statement that gather is not the aim of the thesis) **(2-3 p.)**
 - We use a GRC (no modification) for A and b -> get nb loads
 - Show the interest of a reduction cache region for c -> REDs / reduce the number of mem accesses / link to SoA (see ASAP) **(2 p.)**
 - We created a probabilistic model of the reduction cache which is based on the following principles ... (presented in detail in appendix n) / advantage of this model / we implemented this model in C / validation of the model with state of the art article on generic cache formalisation / introduce the graphs for the next sections **(1-2 p.)**
 - Show that reduction cache does not handle randomness -> model graph **(1-2 p.)**
 - Show RedC 1reg vs RedC 2reg on banded example **(2-3 p.)**
 - Show best cache dimensions for the dense part (band) **(1 p.)**
 - Show that sparse region (out of band) could use an other region for fine grain density (case of sub bands) -> dimensions **(1-2 p.)**
 - Show that sparse region could use adapted HW because of the isolated elements -> dimensions **(1-2 p.)**
 - Show how it would work with a uniform matrix -> how does it affects the dimensions **(1 p.)**
 - Conclude on our solution and dimensioning **(1 p.)**
- Micro architecture of SpDCache **(~10 p.)**
 - Present our hardware solution, based on the concept
 - Start with basic info on HPDCache (only the one that are useful for SpDCache)
 - Describe modif in pipeline / hazard management / throughput
 - Modif in RAM accesses ...
 - Stats: git / bugs / lines of code / synthesis
- RTL simulation results **(~10 p.)**
 - Experimentations: RTL simulation results -> matrices chosen, generated / metrics / ...
 - Results / comparison with theory / analysis / conclusion

- Future enhancement (**~8 p.**)
 - Architecture presentation with three regions (add OBRC, CB...) (**2 p.**)
 - Architecture of eviction anticipation (**2 p.**)
 - Present multicore version -> scale (**2 p.**)
 - Present additional optimisation (region config, learn from first iteration) (**1-2 p.**)
- Conclusions (**~2 p.**)
 - ..
- Appendices
 - All sparse formats (**~5 p.**)
 - Other accelerators of SoA (**~10 p.**)
 - Probabilistic model of the reduction cache (**~10 p.**)
 - Python model (**~1 p.**)

Chapter 2

Background

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From SoA paper

2.1 Sparse Matrix Formats and Algorithms

2.1.1 Sparse Formats

In the absence of a compressed format, matrices are stored in fixed size two-dimensional arrays. With this *dense* format, elements can be randomly accessed ($\mathcal{O}(1)$), and sequential accesses are highly efficient. The two dimensions are called *ranks* which correspond to the rows and columns. For matrices, there are thus two variants of the dense format, depending on which *rank* is stored sequentially in memory. In a *row-wise* (or *row-major*) storage, the elements within the rows are sequential, and conversely for *column-wise* (or *column-major*).

Sparse matrices contain a very large number of zeros, and to avoid storing these repeated zeros, there exist many compressed formats [10, 12, 18, 47, 71, 84], where certain formats are best-suited for matrices with a specific structure. The main principle of these *sparse formats* is to avoid storing the zero elements, compensating with additional *metadata* which gives the position of the non-zero elements. We can cite some of the most used sparse formats like *COO* (*COOrdinate*) that uses three tables of size NNZ with only the non-zero elements (*Payload* table in Figure 2.1b) and their row and column indices (*Row* and *Col IDX* tables), or *CSR* (*Compressed Sparse Row*) (and its *column-major* counterpart *CSC*) that stores pointers to row (or column) positions in the third table instead of indices (*Row PTR* table in Figure 2.1c).

A visual representation of compressed matrices, called *FiberTree* [83], is helpful to understand these *sparse formats*. Each *rank* of a matrix represents a level of a tree where

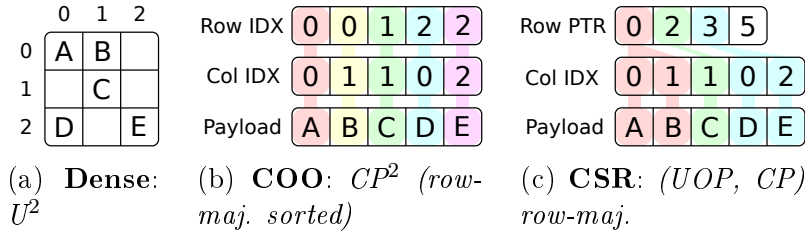


Figure 2.1: Example of a Sparse Matrix in *COO* and *CSR* Formats

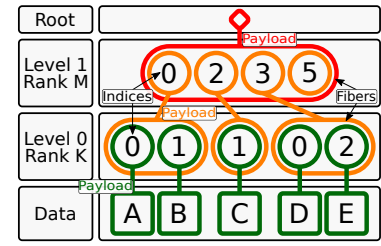


Figure 2.2: Example of the *FiberTree* Representation with *CSR* Format

the nodes are filled with either the *metadata* of the child nodes or the matrix data which can only appear in the leaves (see Figure 2.2). With this representation, we clearly identify the number of *ranks*, which define the *fibers*. A *fiber* is the generic term to describe a one-dimensional stream of data or *metadata*, which corresponds to a row or column in the case of a two-dimensional matrix. A stream of indices is a *fiber* of *metadata*. Figure 2.2 presents, as an example, a *FiberTree* representation of the matrix of Figure 2.1a, matching the *CSR* format: each *rank* in the tree corresponds to a *fiber* stored in the table *Row PTR* or *Col IDX* of Figure 2.1c.

In a recent work [87], a systematic nomenclature for *sparse matrix (and tensor) formats* was proposed, based on *FiberTree* representation. In fact, in the example of Figure 2.2, the *fibers* are compressed with different methods. Thus each *rank* can be characterized by a label indicating how it is compressed: *Uncompressed* (U), *Coordinate Payload* (CP), *Uncompressed Offset Pairs* (UOP). U defines *ranks* where all data values are stored contiguously in memory. CP defines *ranks* that store the indices of each payload, a payload being either a non-zero element or a sub-matrix¹. UOP defines *ranks* that store pointers (offsets) to delimit sub-matrices of the next *rank*². These are the three main ones as they are used in the most common *sparse formats* but many others exist or can be created.

With this nomenclature, the *sparse formats* can be classified according to their *rank* compression methods. For example, the *CSR* format is denoted as (UOP , CP) *row-major* since its *Row PTR* table stores pointers (offsets) to rows. The *CSR* and *CSC* formats are both of type (UOP , CP), one being *row-major*, the other *column-major*. The *COO* format is denoted as CP^2 because there are two coordinates (row and column) for each non-zero element, as it is a CP *fiber* having two indices per data element instead of one. User defined formats can also be easily described and named. For example, a user could define a storage format with a nested *CSR*. That is, each non-zero element in the outer *CSR*, corresponds to a sub-matrix, itself stored in *CSR* format. Such a representation would be labeled (UOP , CP , UOP , CP). A similar nomenclature is proposed by TACO [18]. It is more adapted to code generation applications, where the *FiberTree*-based nomenclature better describes the memory storage of the *sparse format*. For example, U corresponds to *Dense*, CP to *Singleton*, the pair (UOP , CP) is translated as *Compressed* while UOP alone corresponds to *Offset*. We chose to adopt the *FiberTree*-based nomenclature in this article since it is more suited to a hardware context.

¹In this section, we use the term sub-matrix but this is extended to the notion of *tiling* in Section 2.1.3.

²If some sub-matrices are empty, pointers will be repeated, hence 'uncompressed'.

2.1.2 Algorithms

The accelerators described in this paper address sparse MV and MM. For MV we use the notation $A(M, K) \times b(K) = c(M)$ and for MM the notation $A(M, K) \times B(K, N) = C(M, N)$. The *rank* K is the *common rank* between the two inputs.

The existence of different *ranks* allows several algorithmic possibilities to compute matrix multiplication: the *inner-product* algorithm (IP), the *outer-product* algorithm (OP) and *Gustavson's* algorithm (Gust) [26, 35]. In all cases, the matrix A is traversed sequentially. The three algorithms, however, impose different access patterns for B (or b) and C (or c). Indeed, in many cases, they have to be accessed multiple times, as discussed in the following sections.

2.1.2.1 Inner-Product Algorithm

The *inner-product* algorithm [26] is an intuitive way to calculate a matrix multiplication because it follows the loop order of the mathematical formulas (2.1) and (2.2) for MV and MM, respectively.

$$\forall m \in [0, M[, c_m = \sum_{k=0}^{K-1} A_{m,k} \times b_k \quad \forall (m, n) \in [0, M[\times [0, N[, C_{m,n} = \sum_{k=0}^{K-1} A_{m,k} \times B_{k,n} \quad (2.1) \quad (2.2)$$

The *common rank* K is the inner-most loop of the algorithm (see Algorithm 2.1). The output C (or c) is updated sequentially, but the B -matrix (or b -vector) will be entirely read M times. In the context of sparse matrices, the reads of B (or b) are conditioned by the non-zero elements of A , which requires indirect accesses. If B (or b) is also sparse, the algorithm needs to perform *intersections* between the non-zero elements in the rows of A and the columns of B (or in the b -vector) (see Section 2.2). In Algorithm 2.1, exchanging M and N simply causes the output to be updated *column-wise*.

2.1.2.2 Outer-Product Algorithm

The *outer-product* algorithm [26] has the opposite *rank* loop order than the *inner-product*. It changes nothing mathematically except that the computation is separated into two phases. First, during the *multiply* phase, we compute several *partial outputs* (POs) consisting of the *partial sums* $A_{m,k} \times b_k$, then during the *merge* phase, we accumulate all the POs to form the final C -matrix (or c -vector). Algorithm 2.2 show that the *common rank* K is the outer-most loop. For MM, exchanging M and N affects whether the output C is updated by rows or by columns, but has no influence on the indirect accesses to the inputs. With *outer-product*, the B -matrix (or b -vector) is read sequentially. This algorithm necessitates the generation of K POs . Moreover, in the context of sparse input matrices, the PO element updates are conditioned by the non-zeros elements of both inputs. Thus, a *partial sum* could potentially be zero, reducing the number of PO element updates, and making them non-sequential. If the output C (or c) is stored in a *sparse format*, then it is necessary to repeatedly access and update each *partial sum*.

Algorithm 2.1:MM *Inner-Product*

Algorithm

```

1 for  $m \in [0, M[$  do
2   for  $n \in [0, N[$  do
3     for  $k \in [0, K[$ 
4       do
5          $C_{m,n} += A_{m,k} \times B_{k,n};$ 
6       end
7     end
8   end
9 end

```

Algorithm 2.3:MM *Gustavson's Al-*
gorithm

```

1 for  $m \in [0, M[$  do
2   for  $k \in [0, K[$  do
3     for  $n \in [0, N[$ 
4       do
5          $C_{m,n} += A_{m,k} \times B_{k,n};$ 
6       end
7     end
8   end
9 end

```

Algorithm 2.2:MM *Outer-Product*

Algorithm

```

1 for  $k \in [0, K[$  do
2   for  $m \in [0, M[$  do
3     for  $n \in [0, N[$ 
4       do
5          $C_{m,n} += A_{m,k} \times B_{k,n};$ 
6       end
7     end
8   end
9 end

```

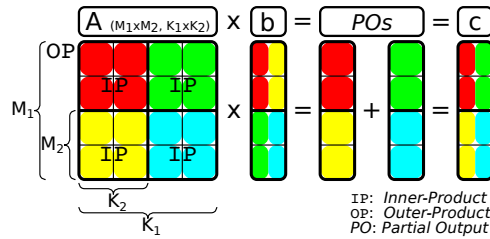


Figure 2.3: An Example of *Tiling* on SpMV

2.1.2.3 Gustavson's Algorithm

As the MM algorithm has three *ranks* instead of two for MV, there is a third ordering, known as *Gustavson's* algorithm [35] where the *common rank* K is the middle loop (see Algorithm 2.3). With this approach, the output is computed row-by-row with *PO*-rows generated and accumulated for each output *C*-row.

With this algorithm, updates of the *C*-rows are sequential, but, in the dense case, the *B*-matrix is entirely read M times and K *PO*-rows of size N are generated. In the context of sparse matrices, the selection of the rows in *B* is indirect, however, within a row the accesses are sequential. The *intersection* operations are only needed for the *B*-rows and not each element. The *PO*-rows can be accumulated before computing the next *C*-row. For this algorithm, exchanging M and N causes the three matrices to be accessed *column-wise* instead of *row-wise*.

2.1.3 Tiling

The *tiling* process [33] consists in matrix partitioning: a matrix may be divided into *tiles*, non-overlapping sub-matrices. In other words, this method consists of adding *ranks* to delimit the *tiles*, and thus adding loop iterations in the kernel algorithm.

Tiling methods allow to locally reduce the dimensions of data and thus to adapt the working set to the size of the local memory [51], particularly when the matrix has a structure with denser regions. Tiling can also increase parallelism opportunities. Software optimizations for sparse matrix computation, such as OSKI [85], are based on *tiling*. In previous sections, the study of the IP, OP and Gust algorithms highlighted the influence of *rank* ordering on the sequentiality and data movement, and thus, adding intermediate *ranks* with *tiling* breaks the regularity of pure IP or OP algorithms, requiring additional hardware support (see Section 2.2).

For example, let us consider the *A*-matrix in Figure 2.3 which is divided into four tiles (one level of *tiling*). For a SpMV, two new *ranks* appear for a total of four *ranks*: M_1 and K_1 allow *tile* selection and M_2 and K_2 allow data access within a *tile*. Thus, we could decide to apply an OP on *tiles* and conversely an IP within *tiles*, mixing-up the pros and cons of these two algorithms at different *rank* levels. In the case of the example in Figure 2.3, the upper-left *A*-tile (red) is evaluated first, sequentially updating the upper tile of the *c*-vector in an IP manner. The lower-left *A*-tile (yellow) is then computed the same way. At this point, *intersections* have been done with only the upper *b*-tile, and *c* has been updated sequentially. The two next *A*-tiles (green, then blue) will compute *intersections* with the lower *b*-tile and update the entire *c*-vector sequentially, showing the OP characteristic of having *POs* to *merge*, two in this case.

Note that in case of *tiling*, the *A*-matrix is no longer read sequentially, which can be problematic if it is stored in a classical *sparse format* like *CSR* that is specifically made

for *row-wise* accesses. Custom *sparse formats* adapted for *tiling* can be used [40], but this requires a preprocessing conversion.

2.2 Hardware Challenges for Sparse Matrices and Vectors

In the previous section we presented a brief background on matrix multiplication without considering the underlying hardware. Although the various *sparse formats* reduce the memory footprint, they have the disadvantage that indirect accesses are required, which creates data-dependencies and reduces the effectiveness of caches. As we have seen, with all three algorithms, at least some of the accesses are non-sequential. The lack of spatial locality for these accesses reduces the benefit of having caches. In other words, a full cache line is loaded, but only a few entries are read or modified. Moreover, the matrices of interest are extremely large and even the non-zero elements, and associated *metadata*, can not generally fit in the cache. Depending on the algorithm, successive accesses to the same element may be too far apart to benefit from temporal locality in the cache.

In an IP algorithm, the *A-metadata* needs to be read to indirectly access the *B*-matrix (or *b*-vector), which may itself be stored in a *sparse format*. In this case, the algorithm needs to check if the non-zero *A*-indices are present in *B* (or *b*). If an index appears in neither list, there is no need to perform the multiplication (*ineffectual operation*). More generally, this problem applies to *fibers* and is called the *intersection search*. As a minimum, this requires reading all the *metadata* for *A* and *B* (or *b*), and identifying those present in both.

In an OP algorithm, the inputs are accessed sequentially, but the updates of the output require indirect accesses. The pattern of the accesses, while the output is generated, is non-sequential and determined by the structure of the input. Furthermore, if these *random*³ updates provoke cache misses where only one word within a cache-line is modified, the bandwidth to external memory is poorly utilized. The OP algorithm generates several *partial outputs* (*POs*) that must be accumulated to obtain the final output. We can identify two strategies: *immediate update* and *deferred update*. In the former, the final output is immediately updated as each *partial sum* is generated. If the output is stored using a *sparse format*, either the index being updated does not yet exist, and the new *partial sum* must be inserted. Or, if it exists, a lookup must be done to locate it, the update performed and the result written back. The problem associated with combining the *POs* is called *reduction*.

To avoid the complexities of the indirect accesses with *immediate updates*, the *partial sums* can be stored and the accumulation can be deferred. One such technique is called *output batching* [48]. With this technique, tuples containing the index and the *partial sum* are streamed sequentially to memory. Then, during a second pass, they are read back and the accumulations are performed, an approach which results in sequential memory accesses during the first pass, with the downside that the volume of data transferred to memory is increased. As we will see, hardware accelerators may combine the *immediate* and *deferred update* techniques to optimize memory bandwidth. In Table 2.1, we summarize the previously discussed advantages and hardware challenges for each algorithm, including Gust which presents a trade-off between IP and OP.

³We use the term *random* to mean accesses are irregular and difficult to predict.

Table 2.1: Comparison of Advantages and Challenges with Matrix Multiplication Algorithms

		Inner-product (IP)	Outer-product (OP)	Gustavson (Gust)
Access Count	MV	A : once ; b : $NNZ(A)$; c : once	A : once ; b : once ; c : $NNZ_r(A)$	N/A
	MM	A : N ; B : $NNZ_r(A).M$; C : once	A : once ; B : $NNZ_c(A)$; C : $NNZ_r(A).NNZ_c(B)/K \sim$	A : once ; B : $NNZ_c(A)$; C : $NNZ_r(A).NNZ_c(B)/K \sim$
Sequential Accesses		Accesses to A and C	Accesses to A and B	Accesses to A and C Accesses to B -rows
Input Format		Accesses to A and B benefit from alternate formats (CSR and CSC)		Allows consistent format for A and B
Parellism Opportunity		Possible with <i>tiling</i>	Generation of POs over <i>common rank</i> of A and B	Reading A -rows and updating C -rows
Reuse Opportunity		Potentially B , but limited by its size	An A -column or B -row while computing POs	A set of B -rows, based on non-zero elements of A -rows
Challenges		<i>Intersection</i> search between non-zero elements in A and B	Storage for POs or in-place <i>merge</i> and <i>reduce</i>	<i>Merge</i> and <i>reduce</i> of PO rows

The issues presented with IP and OP can be generalized to two concepts: *gather* and *scatter*. The *gather* problem exists when the inputs are indirectly accessed, and need to be read from memory using *metadata* to present the processor with the correct terms for computing the *partial sums*. Typically, with IP, the *gather* problem is more difficult. The *scatter* problem exists when the order in which the output matrix/vector is generated is not sequential and is typically associated with the OP algorithm. In the case of *tiling*, a hardware accelerator must handle both *gather* and *scatter*, creating a need for hardware support for both *intersection* searches and *reductions*.

2.2.1 SpMSpM Algorithm Analysis

To better understand how the algorithms impact the number and types of memory accesses, we developed, and presented in Figure 2.4, a model based on memory accesses counts. We identify five memory access patterns: (1) those where a matrix is read only once, thus there is no opportunity for reuse, (2) where the accesses are sequential, but a line is read multiple times, (3) where lines are read consecutively, but the accesses within a line are *random*, (4) where the lines are accessed *randomly*, but within a line the accesses are sequential, and (5) where the matrix elements are accessed multiple times, and with a *random* access pattern.

For each of the three algorithms, we counted the number of read and write accesses, based on the density of the input matrices. We plotted the number of memory accesses in logarithmic scale for SpMSpM, based on a square matrix with $M = 10,000$. Indeed, the number of non-zero elements in C depends on the structure of the input matrices and in this model, we have assumed the non-zero entries in the input matrices are uniformly, randomly distributed. This corresponds to a *worst case* for sparse matrices (no spatial and temporal locality). Banded input matrices would, for example, have fewer non-zero elements in C . In each graph, we have separated the accesses into A (bottom), B (middle)

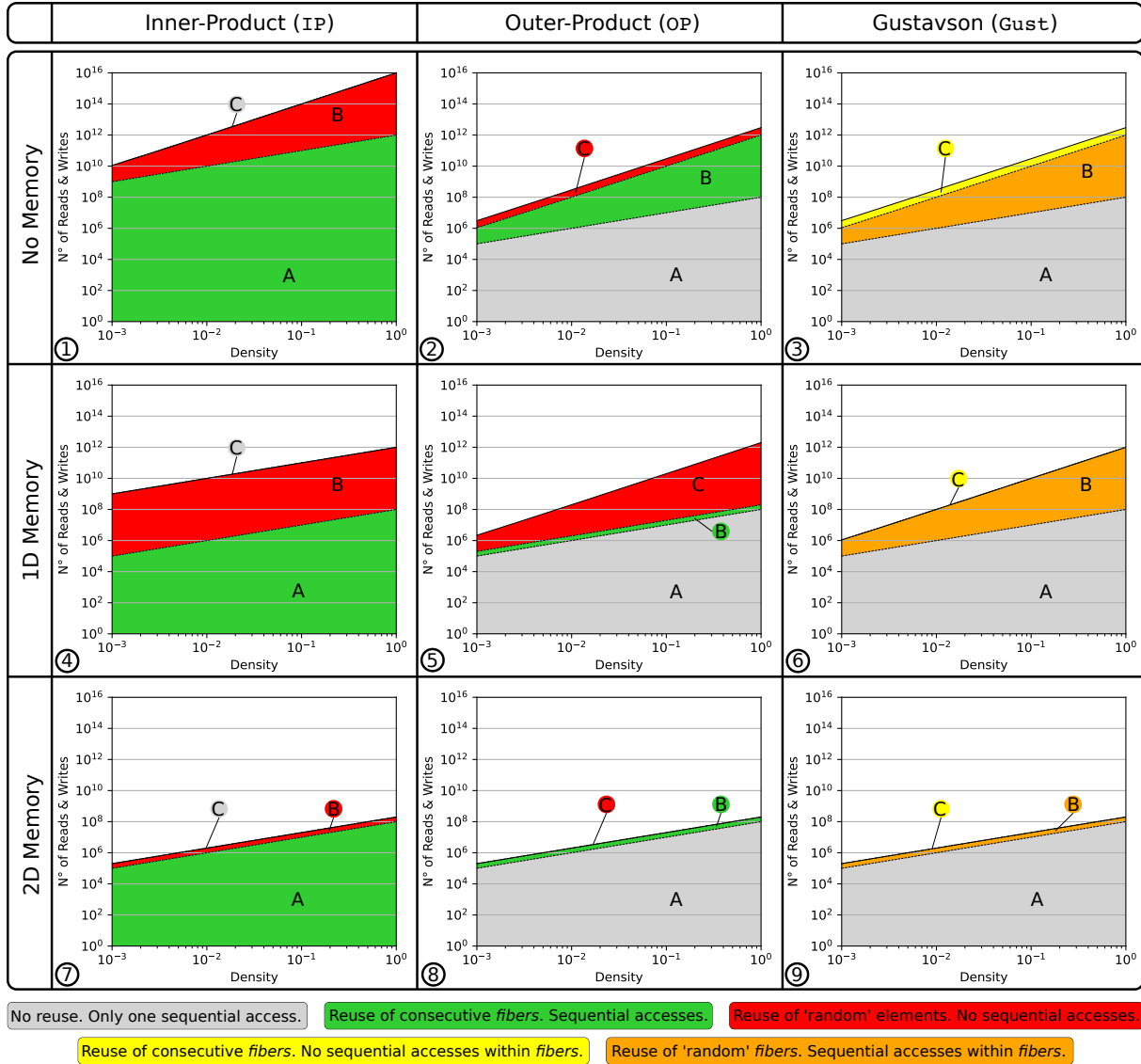


Figure 2.4: Model of Memory Access Counts Depending on Algorithm and Local Memory Size

and C (top). The colour code shows the access pattern, based on the five categories described above.

In the top-row of Figure 2.4 (graphs ①, ② and ③), we start by considering a naïve accelerator which has no local storage, thus all data comes from main memory. From these figures, we clearly see that IP requires more accesses, as the A -matrix must be read N times (graph ①). Whereas for OP and Gust, the A -matrix is read only once (thus the grey colour). For OP and Gust the total number of accesses is the same, however, with OP (graph ②), the accesses to C are *random* (red). With Gust (graph ③), the accesses to B are sequential within the rows (orange) and the C -rows are generated sequentially (yellow), thus Gust avoids having any fully *random* accesses (red).

Since main memory accesses are the principal bottleneck, accelerators integrate local storage (such as buffers or caches). Thus, we consider the case of an accelerator with sufficient local memory to store one full row, per matrix. We assume this local storage is perfect (no misses) and re-calculate the number of remaining main memory accesses, as shown in the middle-row of Figure 2.4 (graphs ④, ⑤ and ⑥). The matrices are assumed

to be initially stored in main memory, thus they must be accessed at least once. As can be seen, for IP (graph ④), the number of A -accesses is reduced, because rows are reused, thus the total number of A -accesses becomes equal to that in OP and Gust. For OP (graph ⑤), the single row buffer greatly reduces the number of B -accesses, however, it does nothing for C (despite the appearance, due to the logarithmic scale). For Gust (graph ⑥), the single row buffer is sufficient to cache the *reduction* operations in C , and thus, it now has fewer overall accesses than OP. It is also interesting to note that OP and Gust scale better than IP with lower densities, as seen by the shallower slope.

Finally, we have considered the extreme case where an accelerator has a buffer of sufficient size to store two matrices, which is shown in the bottom-row of Figure 2.4 (graphs ⑦, ⑧ and ⑨). Although this case is not practical for large matrices, through the proper use of *tiling* to locally reduce the dimensions of a sub-matrix, this case can be achieved at the level of a *tile*. In this case, the total number of external accesses is equal for the three algorithms, as each matrix is accessed only once from main memory. For IP (graph ⑦), it is interesting to note that such a large buffer is needed to locally address the *intersection* search in B . Similarly for OP (graph ⑧), a large storage is required to address the *reductions* in C . For Gust, in fact, it is not necessary to buffer the entire B -matrix. Indeed, buffering a few rows (corresponding to $NNZ_r(A)$) is sufficient to achieve the number of accesses shown in graph ⑨. This corresponds to an intermediate design point (multiple row buffers for B) not explicitly shown in the figure (between graphs ⑥ and ⑨).

When viewed overall, Figure 2.4 highlights the benefits of Gust. We clearly see that Gust does not require any fully *random* accesses (red). With a single row buffer, it is possible to address the *reductions* in C . And, with a limited number of row buffers, the number of external memory accesses for B can be reduced. Based on this analysis, it is clear that Gust is the best algorithm for hardware implementation of SpMSpM.

Chapter 3

State-of-the-Art

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3.1 Introduction

TODO

From SoA paper

3.2 Hardware Accelerators for Sparse Matrices and Vectors

In this section, we describe the recent (2018-2023) hardware accelerators for sparse MM and MV, targeting ASIC-type implementation. We try to be consistent on notations and architecture schemes to simplify the comparison. Before going into the detailed overviews of the accelerators, in Table 3.1 we present a criteria-based classification of the designs.

A first criterion to classify the accelerators is between *standalone* accelerators that address the full kernel without the assist of a processor (*e.g.* OuterSPACE [65] – Section 3.2.1) and ones that address only a part of the kernel and need a processor to manage the computation (*e.g.* PHI [61] – Section 3.2.9). *Standalone* accelerators are directly connected to the main memory and internally manage their own custom memories. The other accelerators actually intervene at different levels of the memory hierarchy, giving a second classification criterion. Certain accelerators intervene near the processors (*e.g.* ASA [91] – Section 3.2.8), near the cache (*e.g.* PHI and SortCache [79] – Sections 3.2.9 and 3.2.7), or near the main memory (*e.g.* SPiDRE [9] and SpaceA [88] – Section 3.2.9).

Each accelerator is optimized for a specific algorithm (IP, OP or Gust), our third criterion. For example, some accelerators (*e.g.* PHI and SortCache) address the *reduction* problem and are adapted to OP algorithms but *reductions* are also required with Gust and highly-*tiled* algorithms. ExTensor [38] (Section 3.2.2) is a general purpose accelerator that is demonstrated for a highly-*tiled* OP but also makes major contributions for IP.

From a memory perspective, the two main problems are *gather* and *scatter*, our fourth criterion. Certain accelerators offer solutions for one, the other or both. For example, some accelerators (*e.g.* SpArch [95] – Section 3.2.3) are specialized for an OP algorithm but optimize both *gather* and *scatter*. Finally, accelerators preferably focus on sparse MM, MV or both – our final criterion.

After this first comparison, in Sections 3.2.1 to 3.2.6 we present the *standalone* accelerators, followed by the *processor assist* accelerators in Sections 3.2.7 to 3.2.8. In Section 3.2.9, we briefly touch on other accelerators.

3.2.1 OuterSPACE

OuterSPACE [65] is an OP accelerator with a focus on SpMSpM kernels, although it can also handle SpMV. It is a highly parallelized architecture which employs Single-Program Multiple-Data (SPMD)-style *processing elements* (*PEs*) that fetch data from main memory through a two-level highly-banked cache hierarchy as shown in Figure 3.1. The input *A*- and *B*-matrices are stored in (*UOP*, *CP*) *column*- and *row-major* formats, respectively, to match the read access patterns of the OP algorithm. The *partial* and final outputs are stored in (*UOP*, *CP*) format.

The computation is temporally divided into the *multiply* and *merge* phases (Figure 3.2). During the first phase, inputs are divided into *tiles* of several rows (*A*) and columns (*B*) and distributed through the *processing tiles* (*PTs*) by a *control unit*. Within each *PT*, each *PE* processes multiplications of one element of the k^{th} *A*-column with all elements of the k^{th} *B*-row and an L0 cache in the *PT* facilitates the reuse of the *B*-rows between *PEs*. This phase generates a *partial output* (*PO*) per *PT*, corresponding to the *tile* of rows (*A*) and columns (*B*) that it was assigned.

Then, during the *merge* phase, the *POs* are combined to generate the final *C*-matrix. Each *PT* locally sorts the *POs*, and then the *PTs* work together to merge them. This algorithm was chosen to maximize data locality and parallelism, although it is less efficient. This *merge* phase only uses half of the *PEs* so the others are disabled (clock-gated) and the caches are reconfigured in a scratchpad mode to save energy.

Note, if *NNZ* elements in the *POs* exceeds the L0 cache capacity, it overflows to the last level cache, which results in reduced performance. OuterSPACE performs best when the $NNZ_c(A)$ and $NNZ_r(B)$ are both uniform which ensures a uniform distribution of work across the large number of *PTs* and *PEs*.

Table 3.1: Algorithmic and Architectural Classification of Existing Accelerators

Name	Date	Autonomy		Position in Memory Hierarchy			Most Adapted Algorithm			Hardware Support for		Sparse Kernels Addressed
		Standalone	Processor Assist	Near Processor	Cache	Near Main Memory	Inner-product	Outer-product	Gustafson	Gather	Scatter	
OuterSPACE [65]	2018					N/A						MV, MM
ExTensor [38]	2019					N/A		a			a	MV ^b , MM
PHI [61]	2019								b			MV, MM
SPiDRE [9]	2019											MV, MM
MatRaptor [81]	2020					N/A						MM
SpArch [95]	2020					N/A						MM
SpaceA [88]	2021											MV
Gamma [92]	2021					N/A					a	MM
InnerSP [6]	2021					N/A						MM
SortCache [79]	2021								b			MV ^b , MM
ASA [91]	2022							b				MV ^b , MM
Flexagon [60]	2023					N/A						MM
GSCSp [55]	2023					N/A						MM

Gray cell: the accelerator meets the criterion.

^aNot main contribution but addressed in the article.

^bCould be adapted but not discussed in the article.

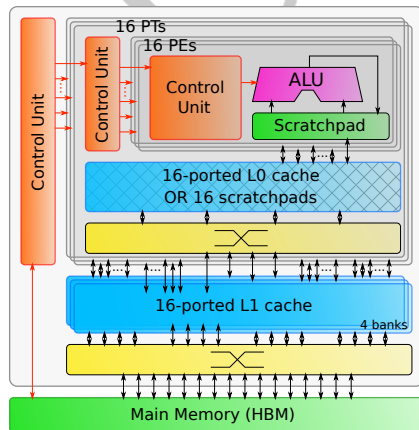


Figure 3.1: Architecture of OuterSPACE

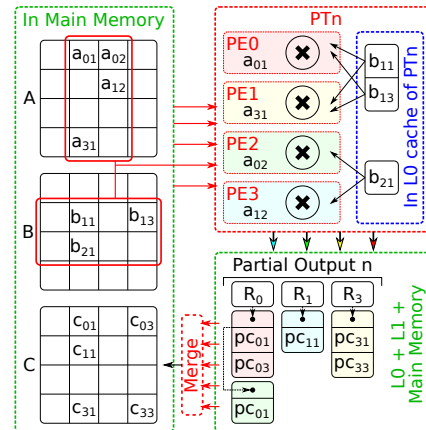


Figure 3.2: OuterSPACE Data Flow

3.2.2 ExTensor

The authors created ExTensor [38] as a general purpose sparse tensor algebra accelerator (including SpMSPM). It is an OP accelerator but due to its support for three levels of fixed *tiling* of the inputs, it optimizes the *intersection* search between non-zero elements or *tiles*¹. By doing the *intersections* ahead of time, at different levels of the memory hierarchy (which correspond to the *tile* levels), ExTensor is able to efficiently eliminate *ineffectual operations*, thus addressing the main issue in IP or complex *tiling* algorithms. Indeed, the input matrices would typically be stored in an (*UOP*, *UOP*, *UOP*, *CP*) format, corresponding to three levels of *tiling*. ExTensor is designed with a flexible architecture composed of different bricks (see Figure 3.3). To simply aid in the understanding, the interactions between the bricks are described using function calls:

- The *Scanner* is a storage unit that delivers a stream of coordinates in increasing order, fetching them from memory. The coordinates are delivered through the *Iterate()* operation call.
- The *Intersect* is a hardware block that co-iterates over several *Scanners* in parallel and compares the coordinate streams. This unit performs the *intersection* search required by IP type algorithms, delivering a single stream with the matching coordinates, which are the coordinates of non-zero *partial sums*. To optimise the *intersection* step, it is possible to skip a range of coordinates in a stream depending on the current coordinate of another stream. Thus, the *Intersect* sends skip messages with *SkipTo()* operations to the other *Scanners*.
- The *Coordinator* is a hardware unit that includes *Scanners* and *Intersect* units, and manages the input and output data depending on the current processed *tile*, as shown in Figure 3.3. Data and *metadata* are respectively stored in storage units and *Scanners*, while a *Sequencer* manages the *Iterate()* calls so that the *Intersect* brick delivers the stream of effective coordinates. In addition, in the *Tile Coord* brick, the offsets of the *tile* are added back, so the output of the *Coordinator* is in absolute coordinates.

These bricks can be placed at different positions in a memory hierarchy, depending on the *tiling* and selected hardware. Thus, the *Coordinator* is able to intersect at coarse-granularity with output data being *metadata* of the next *tile*, skipping an entire *tile* when appropriate, with low computation cost, and also at fine-granularity when output data are the non-zero elements that need to be multiplied.

ExTensor is designed with three levels of *tiling* that each have either input or output sequentiality. First, a *Sequencer* plus *Scanners* block ① is placed close to the main memory to determine which *tiles* to send to the next storage, the *Last Level Buffer (LLB)*. Then, for each *tile*, a *Coordinator* ② in the *LLB* intersects the next level *tiles*, which are sent to several *PEs* that will process in parallel. Finally, in each *PE*, a *Coordinator* ③ intersects the non-zero elements of the computed *tile* and sends them to the arithmetic unit of the *PE* for multiplication.

In addition to these bricks which are the main contributions, ExTensor still needs to perform the *merge* phase, including the *reduction* of the *POs* which are stored in *CP*² format. The *Partial Output Buffer (POB)*, is managed by *tiles*, and the content of the *tile* is stored on-chip using a Content Addressable Memory (CAM) for quick access, enabling *reductions* to be performed. On overflow, the *tiles* are flushed to main memory. To benefit from ExTensor's support for *tiling*, the input matrices need preprocessing, making

¹Tensaurus [82] is another accelerator designed for tensors, but their focus is on SpMM.

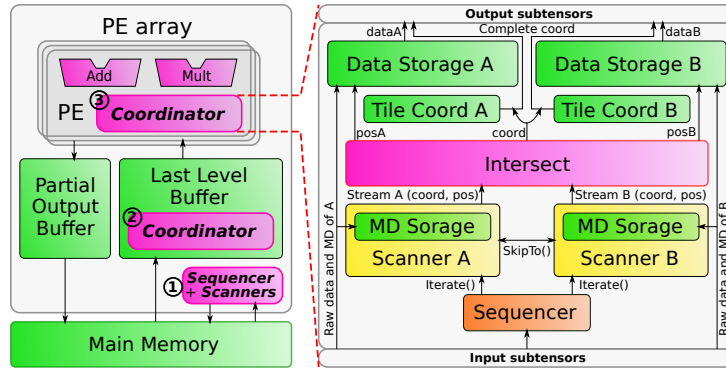


Figure 3.3: Architecture of ExTensor

it difficult to directly compare performance with other accelerators.

3.2.3 SpArch

SpArch [95] uses an OP algorithm for SpGEMM in order to benefit from the simplified access pattern for the inputs. To address the management of the large volume of *POs*, SpArch pipelines the *multiply* and *merge* phases and uses condensing methods to reduce the number of *POs* and scheduling methods to reduce memory traffic. Unlike other OP accelerators [65], with SpArch, both *A* and *B* are stored in (*UOP*, *CP*) *row-major* format. As a consequence of this *CSR* format, the *A*-rows can easily be condensed to the left (see Figure 3.4). Thus, when a condensed *A*-column is read, it will require reading all the *B*-rows corresponding to the different *A*-columns that have been combined. The *MatA Column Fetcher* reads the combined *A*-columns and the *MatB Row Prefetcher* reads the necessary *B*-rows (see Figure 3.5), thus masking the memory latency. The *MatB Row Prefetcher*, stores multiple *B*-rows in a cache, and it uses information about the upcoming *A*-columns (provided by the *Distance List Builder*) to ensure the needed *B*-rows are kept in the cache.

A key contribution of SpArch is the merge unit that is able to merge sixty-four *POs* in parallel, through a binary tree type architecture. The core of the merger is a brick which compares and merges 4×4 (*index*, *value*) pairs in one cycle. These are combined to build a hierarchical merger array whose output is then written to main memory, and read back for further merging by the *Partial Matrix Fetcher*. The authors note that the order in which the *reductions* are performed is important, specifically sparser *POs* should be merged first and they propose a Huffman tree scheduler to determine the best order to minimize memory accesses. Overall, SpArch is able to compute an OP SpGEMM² kernel with reduced main memory traffic for the *POs* mainly by addressing the *scatter* problem with a highly parallelized merger.

3.2.4 MatRaptor

MatRaptor [81] is an accelerator designed to compute SpMSpM kernels. It is based on Gust algorithm³, which makes it possible to process multiple *A*-rows (2 to 8 for MatRaptor) in parallel. In MatRaptor (see Figure 3.6) a *SpAL* (*Sparse Matrix A Loader*) fetches non-zero elements of a *A*-row and sends them to a *SpBL* (*Sparse Matrix B Loader*). With

²Since the merger can read *POs* from memory, the *D*-matrix can be treated like an initial *PO*.

³Called *row-wise product* in [81].

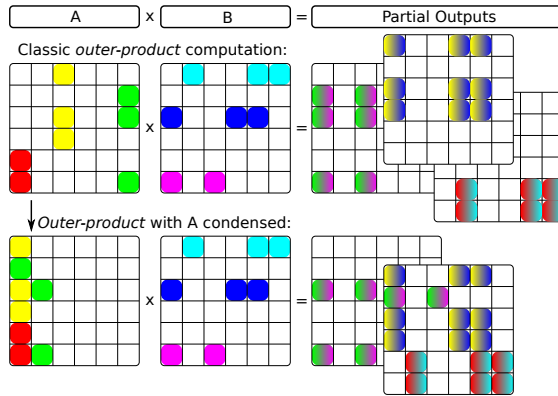


Figure 3.4: *Outer-product* with a Condense *A*-Matrix

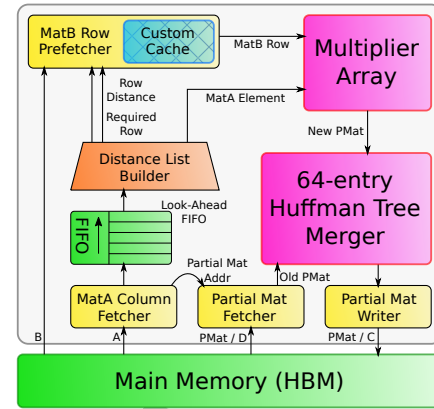


Figure 3.5: Architecture of SpArch

the *metadata* of this element, the *SpBL* fetches non-zero elements of the corresponding *B*-rows and sends pairs of *A*- and *B*-elements to a *PE*. This *PE* computes the *partial sums* and sends the results into queues for the *merge* process. Instead of separating *multiply* and *merge* phases, several queues are used to store *partial sums* and partially merged *partial sums*, allowing the two phases to be pipelined. The *PE* merges all the *partial sums* resulting from an entire *A*-row, before sending the final output (the corresponding *C*-row) to main memory.

In MatRaptor, multiple *A*-rows are processed independently and in parallel in different *PEs* and each *PE* is assigned to a channel in an HBM memory [45]. It is important that each *PE* can read its *A*-rows without interfering with the other *PEs*. With the classic (*UOP*, *CP*) *row-major* format, the rows are consecutive in memory and are not aligned to channel boundaries. To solve this problem, the authors propose a new format called *C²SR* ((*FL*, *UOP*, *CP*) *channel-major*) that is based on a (*UOP*, *CP*) *row-major*. For example, with two *PEs*, all the elements of the even rows are stored in one channel, and the elements of the odd rows, in another channel, thus isolating the data. With this approach, the row pointers (*UOP*) are no longer monotonic, which means that the length of a row can not be deduced by subtracting adjacent row pointers. This requires the addition of a new table (which we call *fiber length FL*), which stores the length of the non-zero elements in each row (see Figure 3.7). This modified storage format enables multiple *PEs* to make effective use of the HBM.

MatRaptor is one of the first accelerators to use Gust algorithm, however, it has some limitations regarding effective reuse of *B*-rows, both between parallel *SpBLs* and temporally, as *A* is traversed. A key contribution is the proposed *C²SR* input format, which makes it possible to separate rows into memory channels, to improve performance.

3.2.5 InnerSP

InnerSP [6] is a SpMSPM accelerator based on the Gust algorithm⁴. The authors of InnerSP justify their choice by showing quantitatively that with the OP the *POs* represent a large volume of data, and that for many matrices the *reductions* can not be done fully on-chip and thus must be buffered in main memory. Normally, with Gust algorithm, *B*

⁴The authors call this *row-wise inner-product*, or confusingly sometimes *inner-product*.

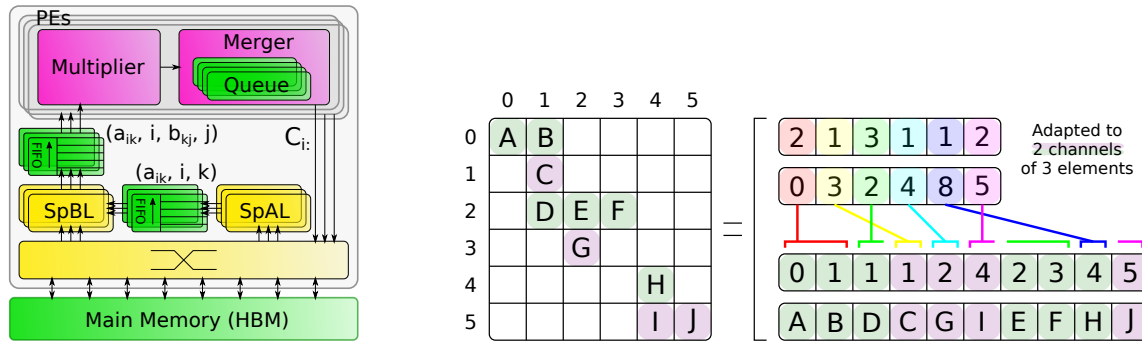


Figure 3.6: Architecture of Figure 3.7: (FL, UOP, CP) Channel-Major Format MatRaptor (C^2SR)

must be read repeatedly, but InnerSP proposes a special cache for B which exploits the spatial locality.

The architecture of InnerSP is shown in Figure 3.8. The A reader reads the A -rows. Two separate caches are used for B , one that stores the row pointers and the other which stores column indices and values. The B reader reads the required B -rows, hopefully getting the data from the caches. Parallel multipliers perform the multiplications and then a hash-unit generates a hash based on the indices in C . A reduction unit based on hash tables contains parallel comparators which search for matching hash values and performs the reductions. When all the operations for a A -row are completed, the C writer writes the C -row back to main memory.

The architecture is efficient, as long as the *Hash Reduction Unit* does not overflow. When overflows occur, the extra entries are temporarily stored in main memory, but there is a significant performance hit. To minimize the chance of overflows, InnerSP uses a pre-scanner to estimate the number of non-zero entries in the upcoming C -rows. The estimate primarily considers the NNZ in the A -rows. If the estimated number of non-zero entries in C exceeds the capacity of the *Hash Reduction Unit*, then the row is split and processed in two passes. Conversely, if the number of non-zero entries in C is small, two A -rows can be processed simultaneously.

To improve the performance of the B -cache, the authors exploit a technique from [7]. The non-zero elements in the A -rows provide direct information about which B -rows are required. The column index table of A is already pre-fetched by the A reader, and when a row must be evicted from the B -cache, it chooses the ones which will not be required based on the upcoming entries in the column index table of A . This technique (also used by SpArch [95]) maximizes the reuse of B -rows.

InnerSP is an accelerator based on Gust algorithm which achieves high reuse of the B -rows, through a look-ahead in the A -metadata.

3.2.6 Gamma

Gamma [92] is a dedicated accelerator for SpMSpM kernels, which uses Gust algorithm. The two input matrices are both stored in (UOP, CP) row-major format and the output is generated in the same format, providing consistency.

The Gamma architecture, presented in Figure 3.9, is composed of a fetcher for the A -rows that contains a *Distributor* to distribute them to 32 PE s. Each PE processes an entire A -row and computes the corresponding C -row. The PE fetches the required

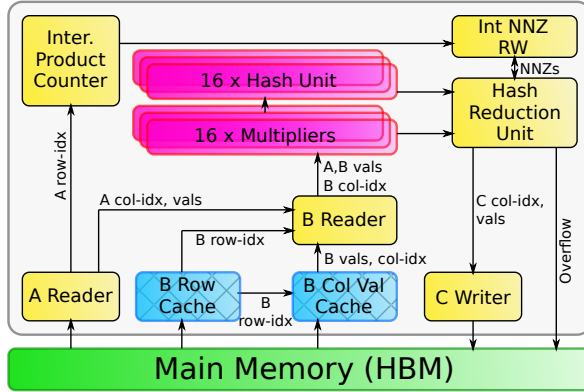


Figure 3.8: Architecture of InnerSP

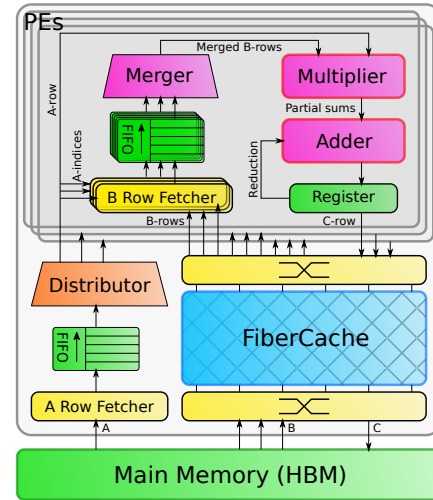


Figure 3.9: Architecture of Gamma

B -rows and merges and sorts them in a 64-wide *merge* block, before multiplying with the A -elements. As a result, the output C -row is generated in order. The *reduction* step is, thus, highly simplified, because any duplicate indices are adjacent. Normally, the C -row can be written back to main memory. However, if the A -row contains more than 64 non-zero elements, then it must be processed with multiple passes. After each pass, the partial C -row is stored temporarily. Finally, the temporary C -rows are read-back and merged by the PEs . Of course, the B -rows must be accessed repeatedly by the different PEs , and Gamma proposes *FiberCache* which is a highly banked cache which fetches the B -rows in advance, based on the column index table of A . It keeps track of which rows of B are most needed by the PEs , and when an eviction is necessary, the victim, is the row which is least needed.

Gamma encounters some difficulties when the A -matrix has rows that are too dense, requiring multiple iterations through the PEs , which creates a high-load on the cache. The authors propose a software preprocessing technique which splits dense A -rows and rearranges them to make best use of the *FiberCache*.

3.2.7 SortCache

The authors of SortCache [79] note that modern processors are able to load an entire cache line in parallel but in many sparse applications less than 50% of the data within a line is used, even with hand-tuned code. They propose a light SpGEMM accelerator which offloads the *reduction* step, based on the *Vectorized Binary Search Tree* (VBST), a data-structure from their earlier work [78, 80]. The VBST is a standard binary search tree where the size of node is a multiple of the cache block size and contains a sorted vector of K (*key, value*) pairs. Associated with each node is a *pivot*; the pivots, with left and right child pointers, form a binary tree, as illustrated in Figure 3.10.

Using the VBST, SortCache focuses on accelerating the *scatter* updates to the output matrix, which are performed directly in the cache. The authors augment the processor with a single new instruction, *RED Key, Value*. After K updates have been issued, the hardware launches a *reduction* in the SortCache, where the K (*key, value*) pairs are inserted into the VBST. This is done by traversing the tree from the root to the leaves.

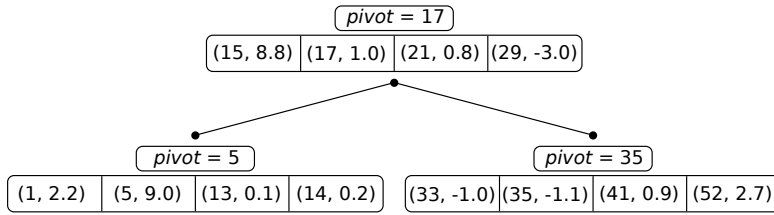


Figure 3.10: VBST Data Structure Used by SortCache

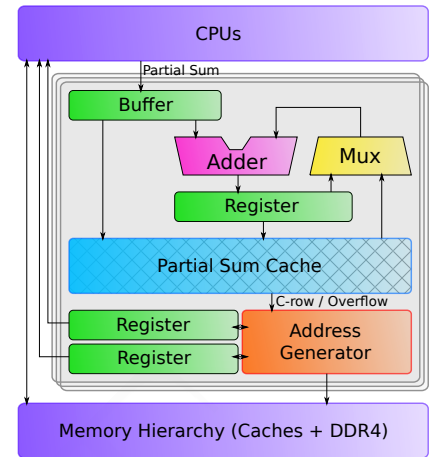


Figure 3.11: Architecture of ASA

When duplicate keys are found, the *reduction* operation is performed in the cache. When new keys are found, they are inserted, which can result in a temporary, new node with up to $2K$ (*key, value*) pairs. This temporary node is partitioned, with one half having exactly K entries. The insertion procedure may require visiting each level of the tree at least once ($\mathcal{O}(\log(n))$) and it does not ensure the uniqueness of the keys, so at the end, an additional full traversal of the tree is required to remove duplicates.

The nodes closest to the root of the tree are accessed most frequently and are thus mapped to the caches closest to the processor. The authors propose to re-purpose cache tracking hardware (e.g. TAG values, LRU counters) to store the additional metadata (pivots, pointers). One of the difficulties with SortCache is that, in some cases, a large fraction of the VBST nodes are underutilized.

3.2.8 ASA

In the scope of graph analytic problems, the authors of ASA [91] observed that Gust algorithm achieves high performance for SpGEMM kernels but note that it still requires acceleration for the *reduction* step⁵. The software state-of-the-art for *reduction* uses hash tables [15]. Some hardware accelerators, such as HTA [94], enhance performance for hash operations, but those solutions are not optimized for SpGEMM kernels. ASA is a low area, near-core extension for a general purpose processor that specifically accelerates the *reduction* step for Gust algorithm.

ASA accelerates *reduction* operations received from software via custom instructions. As presented in Figure 3.11, the *partial sums* from these instructions are stored in a waiting buffer as (*index, value*) pairs with a key that is a hash of the index, computed in software to maintain flexibility. The cache line, accessed with the key, is filled with the indices and the *partial sums*. If a corresponding *partial sum* is already stored, ASA accumulates it and writes back the result. If not, the *partial sum* is simply stored into the cache.

To minimize die area, the cache is not dimensioned for an entire *C-row*⁶ but ASA

⁵The authors refer to the *reduction* step as SPA (Sparse Accumulation).

⁶The authors present a *column-wise* Gust implementation. This choice is arbitrary and, for consis-

handles cache overflows in hardware. When a *partial sum* needs to be stored in a fully filled cache line, a victim is selected and evicted, based on a LRU policy. The evicted entry is written into a pre-allocated FIFO queue in the memory hierarchy. ASA uses an address generator to manage evicted data and updates head and tail registers for the list. Software then traverses the list to merge duplicate keys to build the final output *C*-row. To avoid overflows, the authors advocate *tiling* methods to efficiently distribute work through one or several ASA cores.

ASA is an accelerator which optimizes the *scatter* aspect of SpGEMM kernel with low area overhead, while leaving high software flexibility. Unlike other Gust accelerators, there is no hardware *gather* support for reusing or caching the *B*-rows.

3.2.9 Other Accelerators

In the above sections, we have described a selection of the main sparse accelerators that have appeared in the literature, covering different matrix multiplication algorithms and architectures. This list is not exhaustive, and in the following paragraphs we present several other accelerators which we will cover in less detail, due to space constraints.

3.2.9.1 SPiDRE

The authors of SPiDRE (Sparse Data Rearrange Engine) [9] propose a solution for the poor utilization of the memory hierarchy in sparse applications. They propose a near-memory data rearrangement engine. This engine can preprocess data, near memory, to create new data-structures that are dense and benefit from the memory hierarchy. For example, with a SpMV kernel, the input vector could be rearranged through a user defined function to a dense format where all non-zero elements are in the right order of computation.

3.2.9.2 PHI

The authors of PHI [61] note that many large graph algorithms favour pull (each node reads inputs from its neighbours) compared to push implementations (each node propagates results to its neighbours), as push implementations require read-modify-writes which require accessing complete cache lines, even though a small unit of data is modified. The authors include OP SpMV (a push algorithm) in their evaluation.

PHI is a modified cache optimized to efficiently process commutative *scatter* updates. When an update is received but the line is not in the cache, the line is not fetched, but rather marked as being in *update mode*, where it simply stores the update. If subsequent updates hit the line, they are stored, or combined via the commutative operator. When a cache line is evicted, if it contains multiple updates, PHI assumes that the coalescing has been effective, and it processes the operations via a read-modify-write from main memory. If a cache contains few updates when evicted, PHI batches these updates as (*address, value*) tuples, allocating a new cache line to assemble the batch. Groups of updates are organized by *bins* corresponding to a small memory region. The batches are streamed to main memory. Later, the reading of the batches is also efficient, as they are contiguous.

In multi-core systems, private caches act as local buffers, thus coalescing the updates reduces coherency protocol traffic. This combination of in-cache coalescing and update tency, we stick with a *row-wise* presentation.

buffering enables PHI to take advantage of locality when possible (like Coup [93]), but with the addition of *update batching*, PHI improves memory utilization, even when the data being updated is too large to fit in the cache.

3.2.9.3 GSCSp

GSCSp [55] is a FPGA-based accelerator based on Gust algorithm. In other Gust accelerators, a *PE* processes a full *A*-row but the authors note that when there is variability in the number of elements in the *A*-rows, *PEs* may be blocked, waiting for another *PE* to complete. Thus the authors propose to distribute the elements in an *A*-row to different *PEs* (*element-wise* parallelism). Initially, each *PE* buffers its output *C*-row, and when a *PE* advances to the next row, it pushes its partial *C*-row to a final merger. The authors also propose to use separate caches for the *metadata* for the *B*-rows and the values, and report that combined with the *element-wise* parallelism, the approach is effective for reducing the memory traffic for accessing *B*-rows.

3.2.9.4 Flexagon

Flexagon [60] is a SpMSPM accelerator for DNN applications. The authors claim that the optimal algorithm (IP, OP and Gust) depends on the matrix structure, potentially varying between DNN models and layers within a DNN, although in the majority of the cases, their data shows that Gust is the most efficient. Flexagon is configurable: it can implement all three algorithms. To achieve that, the authors designed a *Merger-Reduction Network (MRN)* that is able to perform multiplications (for *partial sum* generation), accumulations (for *reductions* operations) and comparisons (for *intersection* searches and *merge* phases) via a binary tree structure. Flexagon addresses the memory access issues of each algorithms via three custom memories: a FIFO for the *A*-matrix that is always read sequentially, a cache for the *B*-matrix that is subject to high temporal and spatial reuse in each of the three algorithms, and a PSRAM to store *POs* for OP and Gust algorithms.

3.2.9.5 Other Accelerators

In GraphR [77], the authors propose to use ReRAM technology and a cross-bar to perform graph operations directly in hardware, but this approach is not currently scalable to large problems. The ITS accelerator [72] focuses on an OP implementation of SpMV and they apply data-compression to the *metadata*. The authors of SPU [20] propose a more general hardware solution to input data dependencies and focus on graph applications. The focus of Alrescha [2] is an innovative and adaptive preprocessing of the matrix in hardware to simplify the accesses to the *b*-vector. The authors of CoSparse [27] focus on a software layer that selects the hardware acceleration that is best adapted to the given problem.

The SSSR accelerator [75] for RISC-V processors goes further by addressing *intersection* and *union* while streaming data to and from memory and it can be used for multiple kernels, however the RISC-V performance can not easily be compared with the other accelerators presented in this paper. In passing we note three other accelerators EIE [36], SCNN [66] and CANDLES [34] which have hardware support for sparsity but which are highly focused on CNNs, besides SDMA [30] that focuses on GNNs.

The SpaceA [88] accelerator exploits near-memory computing inside a 3D DRAM memory to implement SpMV using an IP algorithm. With preprocessing, the rows of the matrix are assigned to banks within specific DRAM dies, while optimizing spatial

locality. A separate die is used for storing the input and output vectors. Associated with each DRAM bank which stores the matrix, there is a *PE* which reads the non-zero values of the matrix, fetches the required vector entries and performs the computations. Bringing the compute near the DRAM and exploiting emerging 3D technology, is an orthogonal approach to reduce memory access overheads.

3.2.9.6 Other FPGA based Accelerators

Many sparse accelerators have been prototyped on FPGAs [59]. Indeed, AMD/Xilinx provide a turnkey library for SpMV [19, 56]. In [24], the authors propose a higher-performance SpMV using a modified *CSR* format where the data is packed for efficient, streaming access in HBM⁷. They also propose a highly-banked on-chip memory for the vector updates, and a hazard resolution strategy that allows efficient pipelining.

3.3 Comparison of Existing Accelerators

In this section, we analyze and further compare the accelerators that were described in Section 3.2. We start with Table 3.2 where we summarize the accelerators, chronologically, showing the research group where they were developed. We see that MIT is very active in the field, as they were involved in developing four different accelerators (*e.g.* ExTensor, PHI, SpArch and Gamma).

3.3.1 Benchmarks, Evaluation and Performance

Up to this point, we have discussed the architecture of the various accelerators, without reporting the performance they achieve. Most of the matrices used for benchmarking come from the SuiteSparse Matrix Collection [50] that contains a large number of matrices from diverse, real applications. Domain specific matrix libraries are also used [5, 52, 69, 76], as well as some synthetic matrix generators [1, 11]. The densities are in general less than 1% and can be as low as $10^{-5}\%$. ExTensor, SortCache and Flexagon also benchmark with denser matrices. In all cases, large matrices with several million *NNZ*, exceeding the size of a normal cache, have been evaluated.

In Table 3.3, we note any required preprocessing, assuming, arbitrarily, that matrices are initially stored in *CSC* format⁸. We see that a few accelerators require a conversion to their specific format (*e.g.* ExTensor and MatRaptor). PHI, SortCache and ASA are not concerned by preprocessing since they only address *scatter*. In any case, fair benchmarking must take into account the cost of specialised preprocessing.

Not all accelerators have reached the same level of design maturity, and in Table 3.3, we have shown how each design was analyzed or simulated. All designs report that they have been simulated at a cycle-accurate level. Most of the accelerators use custom simulators or tools like *gem5* [58], *ZSim* [73] and *STONNE* [62], but unfortunately there is no common approach, making it difficult to fairly compare the reported performance. RTL descriptions are often only generated for key parts of the design. To estimate power and area, authors used tools such as *CACTI* [8] and *McPAT* [54]⁹. In the last column

⁷GraphLily [39] is a precursor of HiSparse, also from Cornell University.

⁸This choice is based on the SuiteSparse Matrix Collection where matrices are stored *column-major*.

⁹We have not attempted to compare power/energy as the data in the literature is heterogeneous and can not be fairly compared.

Table 3.2: Chronological Summary of Sparse Accelerators

Name	Date	Institute / University	Summary
OuterSPACE [65]	2018	Univ. of Michigan / Arizona State Univ.	Highly parallelized SPMD, two levels cache – OP SpMSPM
ExTensor [38]	2019	Univ. of Illinois / NVIDIA / MIT	General purpose tensor algebra – With <i>tiled</i> SpMSPM
PHI [61]	2019	MIT (CSAIL) / Carnegie Mellon Univ.	In cache, hash-based accelerator for <i>reductions</i>
SPiDRE [9]	2019	UPC, BSC (Spain) / Arm Research	Near main memory data reorganisation for <i>intersections</i>
MatRaptor [81]	2020	Cornell	Row-parallel with HBM-aware custom format – Gust SpMSPM
SpArch [95]	2020	MIT / Stanford / NVIDIA	Compression method to reduce <i>POs</i> , <i>multiply/merge</i> pipe – OP SpGEMM
SpaceA [88]	2021	UC Santa Barbara	In memory accelerator for 3D DRAM – IP SpMV
Gamma [92]	2021	MIT (CSAIL)	Parallelized architecture with <i>FiberCache</i> for <i>B</i> – Gust SpMSPM
InnerSP [6]	2021	KAIST / DGIST (S. Korea)	<i>A</i> -look-ahead, caches for <i>B</i> , <i>Hash Reduction Unit</i> for <i>C</i> – Gust SpMSPM
SortCache [79]	2021	Georgia Tech / Zettaflops / Sandia Labs	In cache accelerator for <i>reductions</i> , based on the VBST
ASA [91]	2022	Lehigh Univ. / Lawrence Berkeley	Custom cache accelerator for <i>reductions</i> – Gust SpGEMM
Flexagon [60]	2023	Univ. of Murcia / Georgia Tech / NVIDIA	Configurable accelerator supporting all three algorithms – SpMSPM
GSCSp [55]	2023	Nanyang Technological Univ.	Element-wise, FPGA-based accelerator – Gust SpMSPM

Table 3.3: Qualitative Analysis and Evaluation of Sparse Accelerators

Name	Preprocessing ¹	Functional Evaluation (Tool / Evaluation Specifics / Host ²)	Local Memory Size
OuterSPACE [65]	$A \rightarrow$ <i>row-major</i>	<i>gem5</i> / Model only key elements; Skip memory allocation; Ignore start-up time / $\emptyset$	528KiB (33.8k entries)
ExTensor [38]	$A, B \rightarrow$ highly <i>tiled</i> custom format	Custom Python simulator / Only the <i>Coordinator</i> is evaluated; Remainder modeled analytically / $\emptyset$	~38MiB (2.5M entries)
PHI [61]	<i>N/A</i>	<i>ZSim</i> / Entirely evaluated / 16-cores system with 3 level cache	32.3MiB (2.8M entries)
SPiDRE [9]	Not needed	<i>gem5</i> / <i>Unknown</i> / Single ARM core	$\emptyset$ (main memory size)
MatRaptor [81]	$A, B \rightarrow$ C^2SR	<i>gem5</i> / Entirely evaluated / $\emptyset$	320KiB (27.3k entries)
SpArch [95]	$A, B \rightarrow$ <i>row-major</i>	Custom C++ simulator / Entirely evaluated / $\emptyset$	940KiB (62.5k entries)
SpaceA [88]	$A \rightarrow$ row-aligned to DRAM banks	Custom simulator / Simulation tracking values stored in 3D-DRAM / $\emptyset$	4GiB (268.4M entries)
Gamma [92]	$A, B \rightarrow$ <i>row-major</i>	Custom simulator / Entirely evaluated / $\emptyset$	3MiB (262.1k entries)
InnerSP [6]	Not needed	Custom simulator / Entirely evaluated / $\emptyset$	$\leq 976.5\text{KiB}$ (119.9k entries)
SortCache [79]	<i>N/A</i>	Custom simulator / Entirely evaluated / Single threaded, in-order core with 3 level cache	16.3MiB (1.4M entries)
ASA [91]	<i>N/A</i>	Modified <i>ZSim</i> / Entirely evaluated, one ASA per core / 8-cores with 3 level cache	8×8KiB (8×1k entries)
Flexagon [60]	Not needed	Custom simulator based on STONNE / Entirely evaluated / $\emptyset$	1.3MiB (327.7k entries)
GSCSp [55]	$A, B \rightarrow$ <i>row-major</i>	HLS / FPGA Prototype / $\emptyset$	2.1MiB (178.2k entries)

¹We assume matrices are initially stored in (*UOP*, *CP*) *column-major* format (*CSC*).²Only for *processor assist* accelerators.

of the table, we show the size of the local memory storage used in simulation by each accelerator. Except for SPiDRE and SpaceA which work directly in main memory, the local memories are in the range of 0.3MiB to 38MiB, corresponding to 27k to 2.8M entries. Thus, in the model presented in Section 2.2.1, most of the accelerators are situated in the middle-row of Figure 2.4, with a local memory corresponding to a few matrix rows.

The accelerators performance are reported as a speedup versus a baseline, which may differ according to their hardware and software platforms: typically the Intel MKL library [41] and the Armadillo library [74] for CPUs, or cuSPARSE [64] and CUSP [21] for GPUs. Since different baselines with different numbers of threads and cores are used, these speedups need to be interpreted with care. The speedups are in general higher for the *standalone* accelerators as they provide optimized hardware for the entire kernel, while *processor assist* accelerators have less impressive speedups but aim to be general purpose and have lower area overhead.

The only accelerators that have a comparable baseline are the *standalone* ones addressing SpMSPM. They all compare themselves to OuterSPACE with the same benchmark matrices, so we can sort them by their speedup: Gamma ($6.6\times$)¹⁰, InnerSP ($4.57\times$), SpArch ($4\times$), MatRaptor ($1.8\times$) and OuterSPACE ($1\times$). Three of the five *standalone* accelerators use the Gust algorithm (Gamma, InnerSP and MatRaptor). SpArch, an OP accelerator employing a complex architecture, outperforms MatRaptor, but does not achieve the same performance as InnerSP or Gamma, which suggests that Gust is most adapted to high-performance hardware accelerators.

These measurements are consistent with our model presented in Section 2.2.1 (Figure 2.4). OuterSPACE, the baseline, corresponds to graph ⑤. MatRaptor, the first Gust accelerator, was able to achieve higher performance by reducing the number of *random* accesses, and creating better regularity (red to yellow), as seen in graph ⑥.

SpArch, is an OP accelerator with a much larger local memory, which is used to address the red region in graph ⑤. Their *A*-matrix condensing method, allows them to reduce the number of *POs*, and approach the performance achievable in graph ⑧, although, *in fine*, it is impossible to store all the *POs* in local memory. InnerSP and Gamma correspond to optimizations of the Gust approach in graph ⑥, to approach the performance in graph ⑨. This is achieved by custom caches for the *B*-matrix to address the orange region. With the Gust algorithm, it is only necessary to store a limited number of *B*-rows, which is achievable with the large local memories in these accelerators. The model presented in Section 2.2.1 is helpful to understand the local memory requirements for SpMSPM accelerators.

3.3.2 Analysis for Designers

There is no single "best overall" accelerator. A designer must consider the required performance, the available area and the class of problems to be addressed. In Figure 3.12 we present a flow diagram which can guide a designer towards an architecture based on their requirements. Starting from the top-right of the figure, the designer must first decide between a *specialized accelerator* for a specific kernel or a *general purpose accelerator*. The former is preferable if performance must be maximized, requiring a *standalone* accelerator addressing both *gather* and *scatter* (e.g. MatRaptor, SpArch, Gamma and InnerSP).

A *general purpose accelerator* can either take the form of a highly configurable *standalone* block (e.g. OuterSPACE and ExTensor) or a *processor assist* that boosts perfor-

¹⁰This speedup increases to $7.7\times$ with their proposed preprocessing.

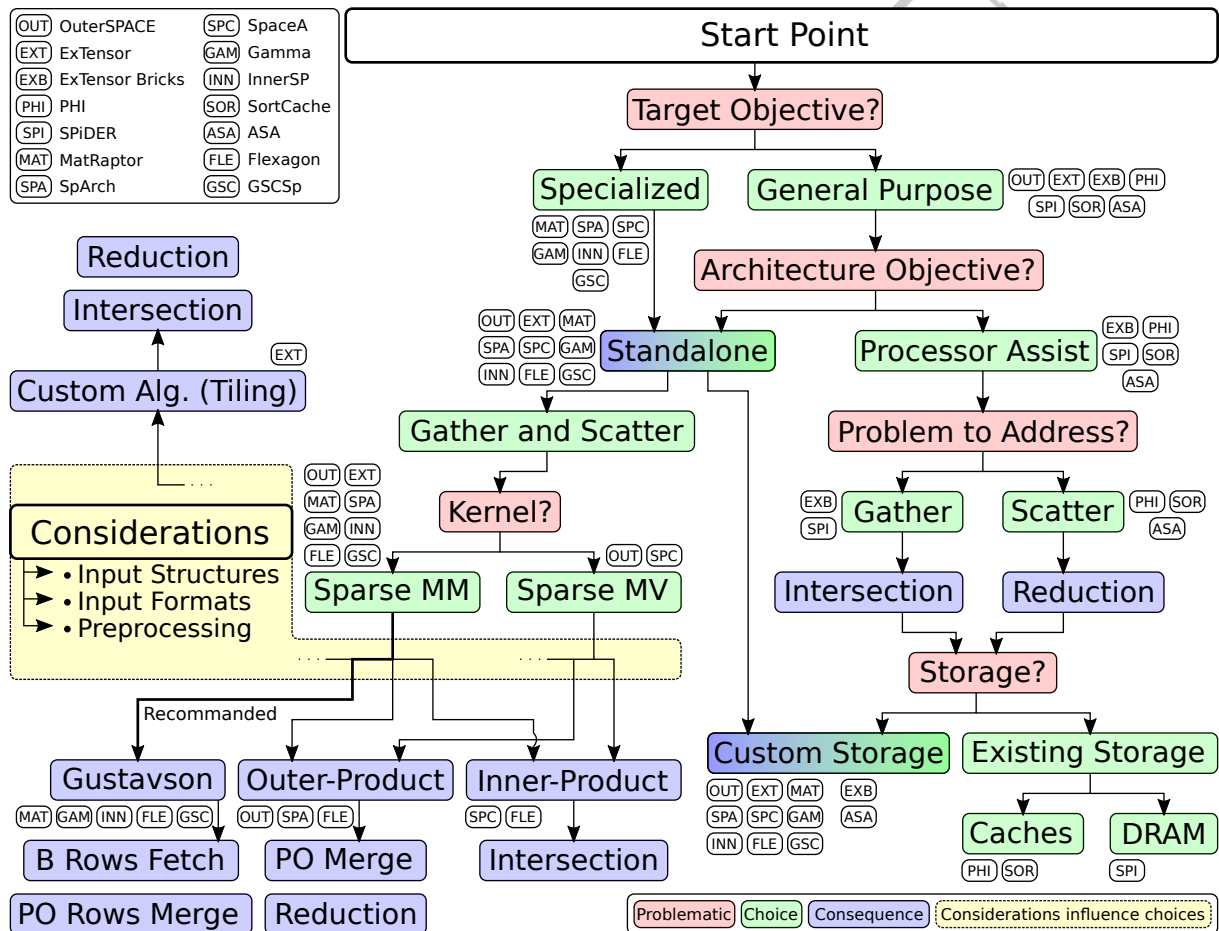


Figure 3.12: Flow Diagram for Architecture Selection

mance for a specific task. With a *processor assist* (rightmost branch in the figure), the processor still performs important parts of the kernel, typically the multiplications, but off-loads the *scatter* or *gather* tasks to the accelerator, as these are the tasks which are the bottleneck with a traditional cache hierarchy. For example, for an IP algorithm, the difficult part of the *gather* task is performing the *intersection* (e.g. the ExTensor bricks and SPiDRE). Conversely, with OP or Gust, the challenge with the *scatter* is to efficiently perform *reductions* (e.g. PHI, SortCache and ASA). Finally, the designer must decide whether the storage is done by reusing existing memory resources (e.g. PHI, SPiDRE and SortCache) or through custom caches/memories.

Going back to the case of a *standalone* accelerator, the algorithms differ according to the supported kernels. For sparse MV, the accelerators studied here have opted for an OP algorithm. For sparse MM, the designers of three of the five highest performing accelerators have chosen Gust algorithm, as this algorithm provides a good trade-off, simplifying the *gather* by reusing input B -rows and the row-by-row generation of the output, simplifying the *scatter* aspect. The fact that the outputs are generated row-by-row, facilitates parallel implementations. Gust works well with (UOP , CP), providing consistent input and output formats. Beyond the basic algorithm, *tiling* approaches, as proposed by ExTensor, can be explored.

Orthogonal to previous choices, other design considerations are important. First, the starting point needs to be defined: some accelerators require software to perform *tiling* (e.g. ExTensor) or to rearrange the rows to improve performance (e.g. Gamma). The cost of such preprocessing can be high, and is usually only justified if the same matrix is reused multiple times.

3.4 Conclusion

The size of the datasets considered in computational physics, machine learning and graph analytics is continuing to grow and over the last six years, we note the emergence of many hardware accelerators designed to improve upon the performance of CPUs and GPUs. In this paper, we have presented the most important designs and established a comparison, while highlighting the underlying hardware challenges associated with sparse multiplication kernels.

As we have seen, with this class of problem, the challenge is to efficiently use the memory system despite the dereferencing required to access the *metadata* associated with the *sparse data formats*. Depending on the selected MM algorithm, the major challenge is either with the *gather* and *intersection* or with the *scatter* updates. We have identified that accelerators based on *Gustavson's* algorithm achieve a trade-off, where, on the *gather* side, a limited number of B -rows need to be accessed, and on the output side, the result is generated row-by-row, simplifying the *scatter* updates. The best *Gustavson* accelerators have implemented further optimizations such as smart cache eviction strategies where the victim is selected by looking ahead at the A -indices. Several of the smaller *processor assist* accelerators reuse existing cache memories and simply modify the behaviour of the control logic (directories, tags, eviction policies) to optimize the operation for sparse matrices. On the other hand, the highest performance accelerators propose dedicated, massively parallel *reduction/mergers*, to maximize parallelism.

Given the number and diversity of existing designs, it is clear that further hardware optimizations and improvements are possible. As stated earlier, there may never be a single "best" sparse accelerator, but rather designs that are best suited to specific classes

of problems. We believe that the structure of the matrices is critical in terms of the sizing of caches and other resources, and that in the coming years, new accelerators will be increasingly tailored based on the structure of the matrices for specific classes of problems.

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Chapter 4

SpDCache

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Conclusion

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